

Menna: Data Profiling and Preprocessing

Primary Responsibilities: Data cleaning, validation, and preparation for visualization

- Identified data quality issues including missing values and inconsistent column naming
- Removed columns with much nulls (Attractions, FaxNumber, Map, HotelWebsiteUrl, countyCode, cityCode, HotelCode)
- removing leading spaces from columns headers
- Selected most important columns for analysis: countyName, cityName, HotelName, HotelRating, Address, Description, HotelFacilities, PhoneNumber, PinCode
- Dropped rows with missing values, reducing dataset from 1,010,033 to 651,079 rows
- Validated the cleaned dataset schema using Pandera library
- Generated Sweetviz profiling report to document data information
- Saved cleaned dataset to "cleaned_hotels.csv"

Rozina: Visualization and Insights

Primary Responsibilities: generating insights through visualizations

- Created bar chart showing top 10 countries by hotel count
- Created bar chart showing top 10 cities by hotel count
- Built average rating by country chart to identify highest-rated destinations
- Converted text-based HotelRating values to numeric format for proper analysis

- Mapped rating values: FiveStar→5, FourStar→4, ThreeStar→3, TwoStar→2, OneStar→1
- Handled "All" data value by mapping to 5-star rating
- Generated horizontal bar charts for better readability of country and hotel rankings
- Added summary statistics displays showing total hotels, average rating, and unique counts